RF AI 201 · FINAL PROJECT · 2026

A pilot machine-learning model for fuel-requirement estimation in agricultural mechanization.

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A single generalized coefficient cannot describe two machines this far apart.

BAFE FUEL FORMULA
STATIC · 1990S · SINGLE COEFFICIENT

Combine Harvester

≠

TREATED ALIKE

Stationary Water Pump

Two official datasets, joined at record level.

BAFE-AMTEC · 1990–2026
ABEMIS · 17 REGIONS

SOURCE A · LAB

376

AMTEC test records

Standardized, controlled tests. Fuel consumption · power · field efficiency · machinery class.

→ PIPELINE

ML Feature Engineering

Hierarchical join · derived columns (L/h, L/kWh) · Random Forest · SHAP · LIME.

SOURCE B · FIELD

246,741

ABEMIS inventory records

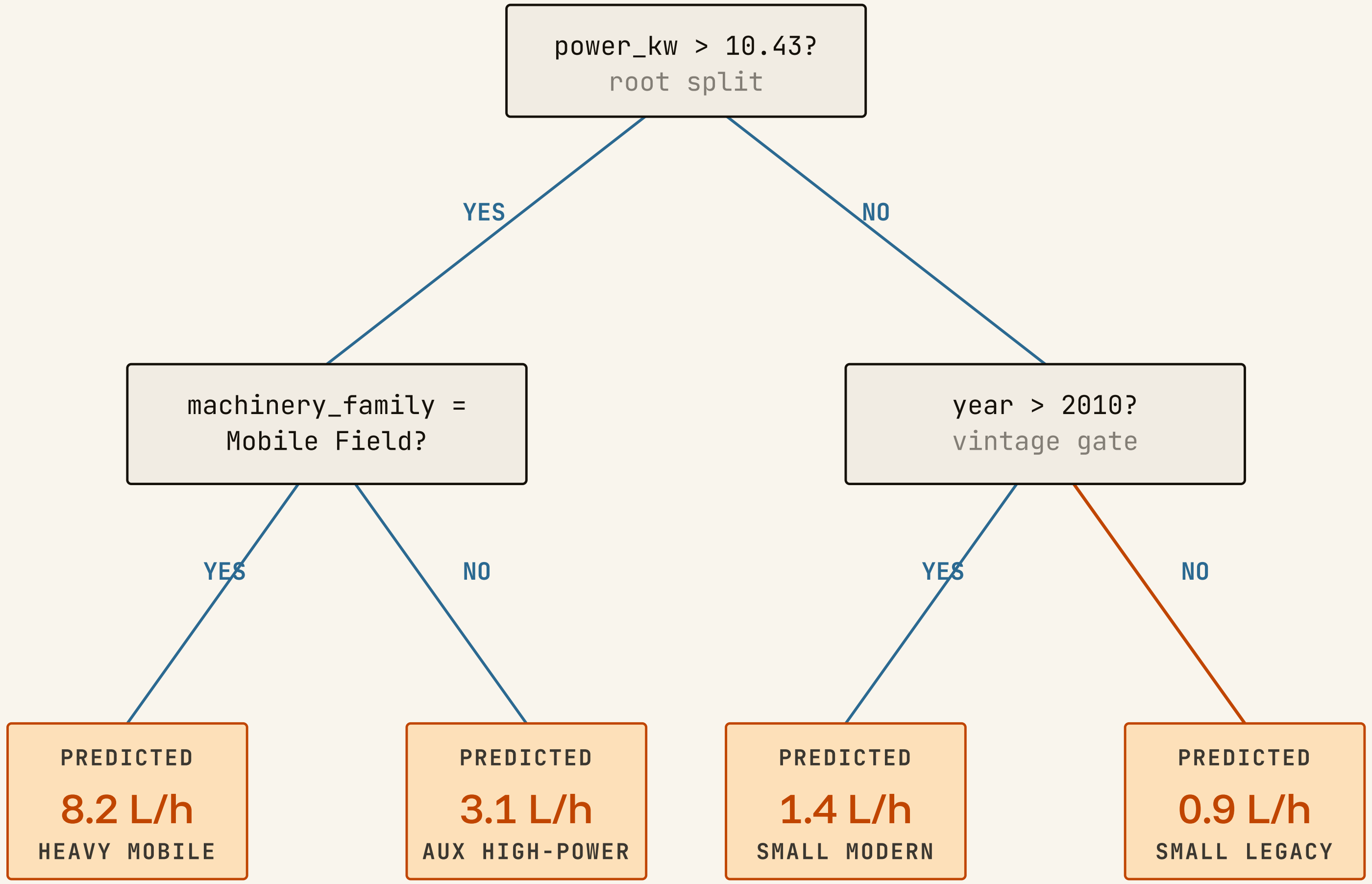
National deployment context: machinery type · region · vintage · operator class.

AMTEC
Bureau of Agricultural and Fisheries Engineering · Agricultural Machinery Testing & Evaluation Center. Standardized lab and field tests since the 1990s.

ABEMIS
National machinery inventory across all 17 Philippine regions. Used here to derive regional prevalence and type distribution as context features.

Power and fuel do not scale linearly across machinery types.

ENSEMBLE · 300 TREES
NO MANUAL FEATURE CROSSING



Why not OLS?

- Power and fuel do not scale linearly across machine types.
- Each tree partitions the feature space differently.
- 100+ trees average out variance, robust predictions.
- No manual feature crossing required.

0.916 R^2 · Random Forest *vs.* ~~0.704~~ R^2 · OLS baseline

Random Forest improves R^2 by 30 % and halves the absolute error.

HOLDOUT · N = 76
80/20 RANDOM SPLIT · SEED = 42

MODEL	R^2	MAE (L/H)	RMSE (L/H)
OLS · Global Linear	0.704	1.882	2.875
OLS · Hierarchical Fallback	0.703	1.879	2.879
Random Forest	**0.916**	**0.950**	**1.534**

+30%
BETTER R^2

2×
LOWER MAE

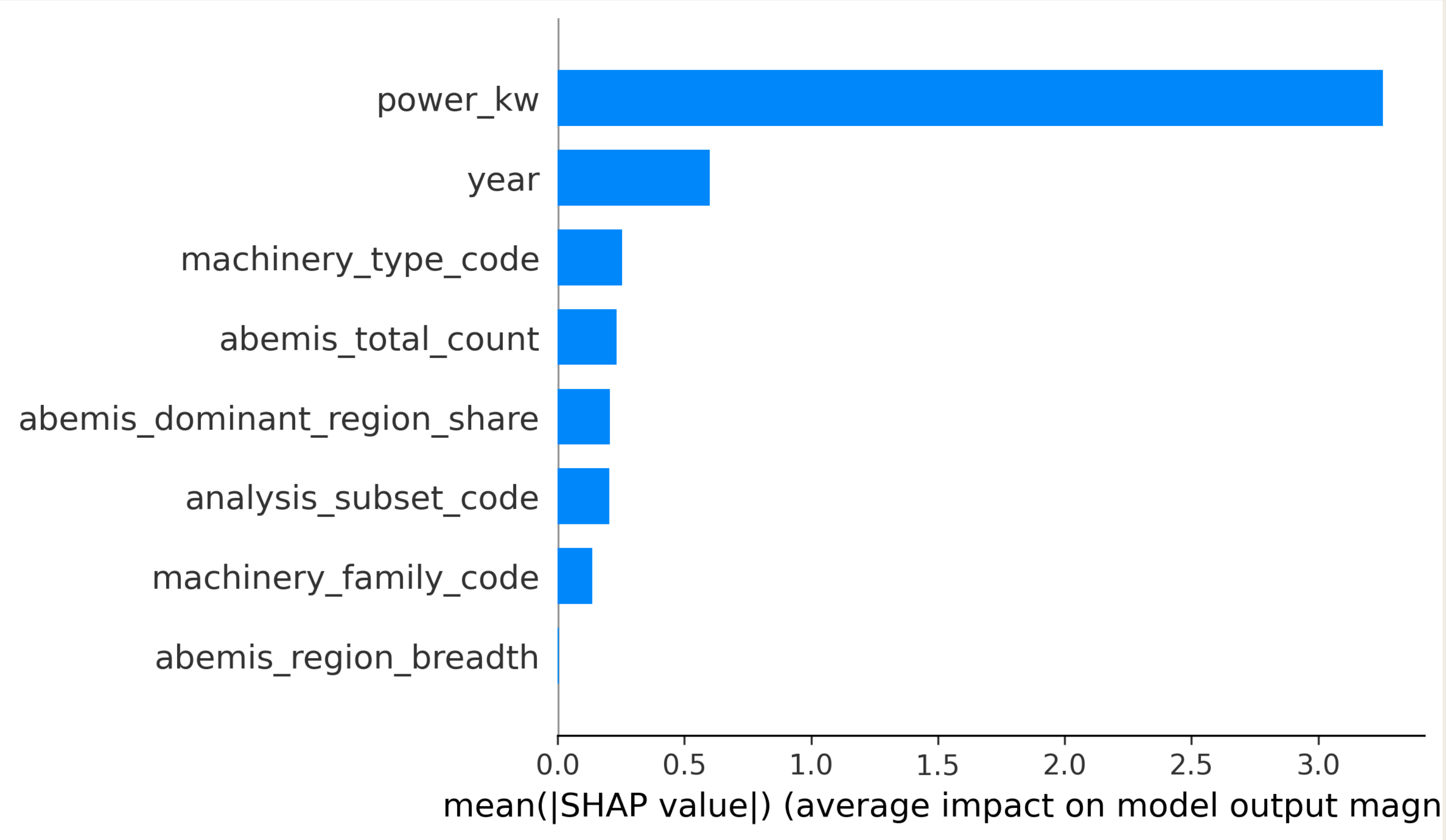
2×
LOWER RMSE

How the model justifies each prediction.

Strong R^2 is not enough. BAFE engineers must understand why the model made each call before they will trust it for national planning. SHAP for the model. LIME for the case.

Global feature attribution recovers the physics of fuel consumption.

SHAP · TREEEXPLAINER
N = 376 · ALL FEATURES



What drives the predictions?

~66.5 %

SHARE OF TOTAL IMPORTANCE · POWER_KW

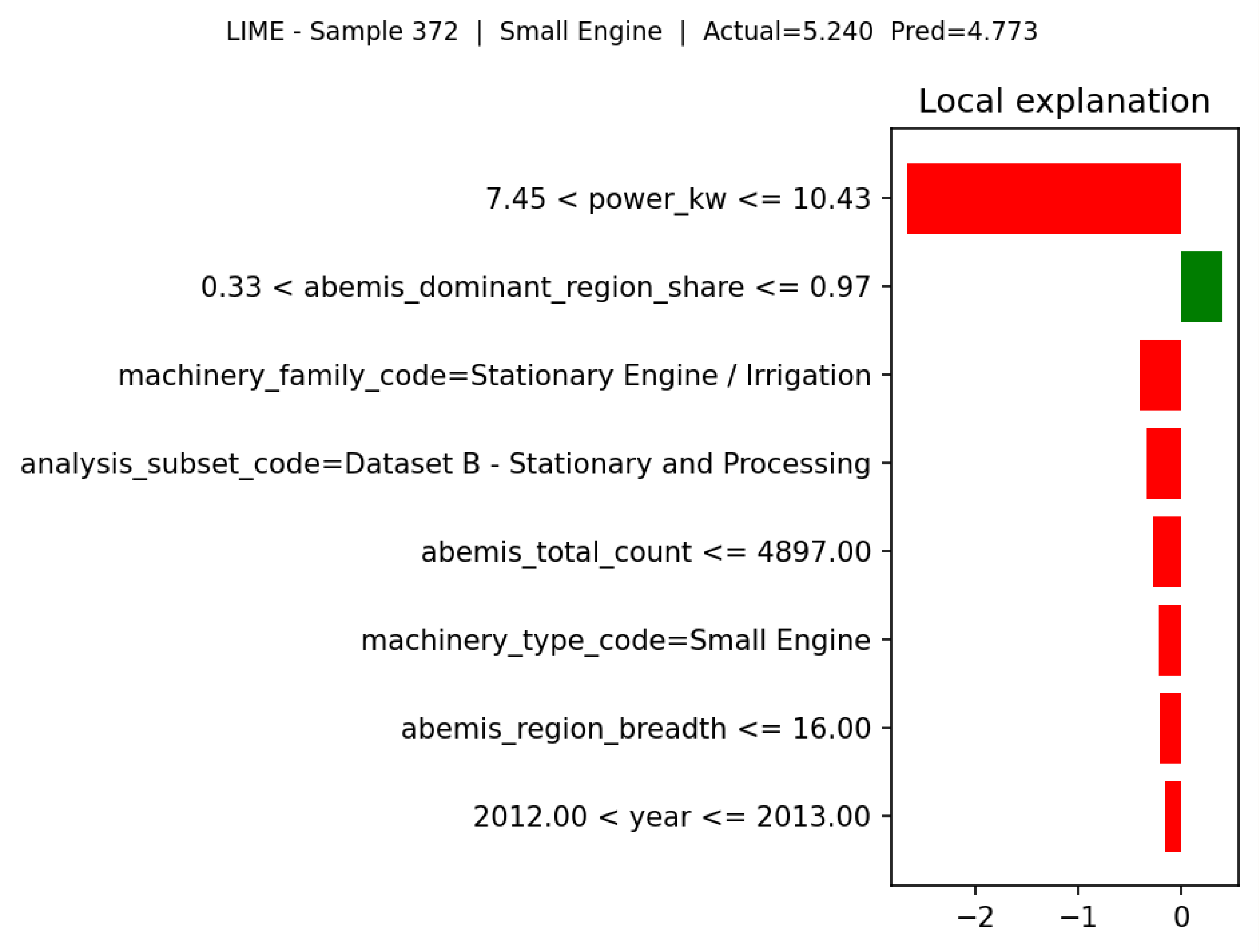
Engine power alone accounts for roughly two-thirds of the model's prediction signal. This directly mirrors the physical law of fuel-power proportionality.

year (equipment vintage) is second at ~12.3%, reflecting improved combustion efficiency in newer equipment. ABEMIS regional features contribute a combined ~9% of contextual fine-tuning.

It learned what an engineer would expect.

Per-instance attribution for a single small engine.

LIME · LINEAR SURROGATE
SINGLE AMTEC INSTANCE



PREDICTED

4.77 L/h

Why this number, for this engine?

LIME fits a local linear surrogate around the prediction and reports per-feature contributions: the rated power pushed the estimate up; the machinery family pulled it down.

This is the granular transparency BAFE engineers need to actually adopt the tool, not global metrics.

Inference applied to the full national inventory.

ABEMIS · 17 REGIONS
RAW INVENTORY: 246,741

172,265

machines successfully scored across the national inventory.

National inventory	246,741
Coverage	70 %
Missing power data	62,808
Unmappable type	11,668

Outputs per region

01 Per-record fuel estimate, in L/h.

02 Per-region aggregate fuel demand.

03 Per-region × machinery-type breakdown.

The unscored 30 % is a data gap, not a model failure. Closing those gaps is on the roadmap.

Closing the data gaps. Validating in the field.

Three concrete next iterations: enlarge the corpus, integrate the cropping calendar, validate against in-field telemetry.

Three iterations from pilot to deployed planning tool.

ROADMAP · 2026 →
BAFE · RAED · FIELD PILOT

01 / Corpus

Expand the *training* corpus.

Acquire the 3,065-PDF collaborator set. Tesseract 5.5 OCR is already deployed on 267 scanned reports, which already brought the corpus from 371 to 376 records (24 OCR candidates, 5 net new after cleaning).

Current: 376 records · Target: ~3,400

02 / Calendar

Integrate the *RAED cropping calendar*.

Adds temporal-load timing as a predictive feature: when does each machinery type actually run, by region and crop stage? Form acquired; data not yet ingested.

Adds: is_in_season · crop_stage_intensity

03 / Telemetry

Validate with *field telemetry*.

Pilot deployment with GPS and flow-sensor instrumentation on a regional machinery cohort. Closes the lab-to-field loop the AMTEC corpus alone cannot.

Validation: in-field, instrumented

Thank you.

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